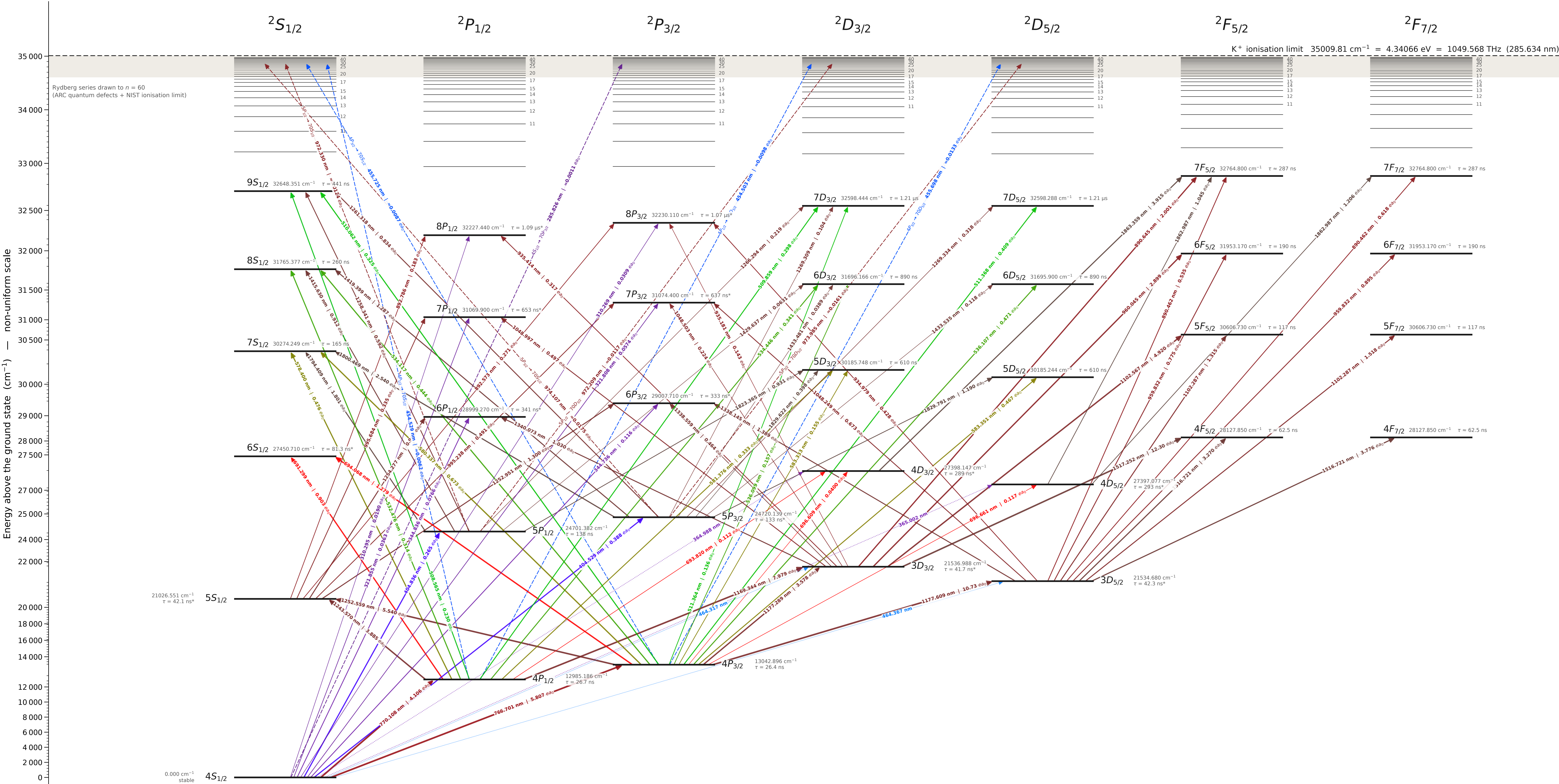
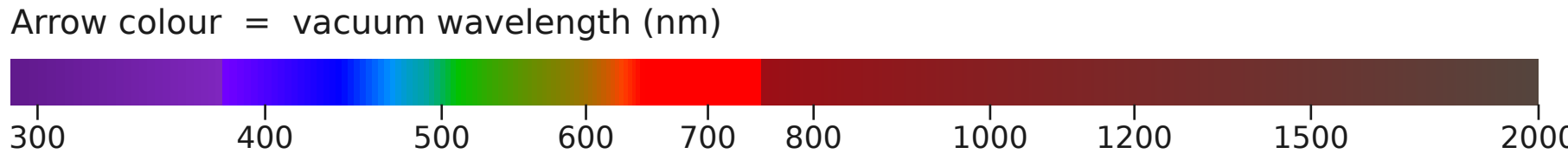


Potassium-40

⁴⁰K (K I), nuclear spin *I* = 4. Energy levels and transitions below 2 μm: 34 levels, 84 lines.

Arrow label: vacuum wavelength | reduced dipole matrix element ($| \langle J || e r || J' \rangle |$) in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked ≈).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\tilde{\nu}$ (cm ⁻¹)	$ \langle J e r J' \rangle $ (ea0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Typical use
4S _{1/2} – 4P _{3/2}	766.7007	766.4897	391.0163	13042.900	5.807	3.790e+07	6.031	100.0	D2 line
4S _{1/2} – 4P _{1/2}	770.1081	769.8962	389.2862	12985.189	4.106	3.740e+07	5.952	100.0	D1 line
4P _{3/2} – 3D _{5/2}	1177.6089	1177.2866	254.5773	8491.784	10.73	2.383e+07	3.764	100.0	
4P _{1/2} – 3D _{3/2}	1169.3442	1169.0241	256.3766	8551.802	7.979	2.017e+07	3.814	83.5	
4P _{3/2} – 5S _{1/2}	1252.5591	1252.2165	239.3440	7983.655	5.540	1.582e+07	3.780	66.5	
4P _{1/2} – 5S _{1/2}	1243.5699	1243.2298	241.0741	8041.365	3.885	7.951e+06	3.780	33.5	
4P _{3/2} – 6S _{1/2}	694.0678	693.8764	431.9354	14407.814	1.279	4.956e+06	1.959	40.3	
4P _{3/2} – 3D _{3/2}	1177.2889	1176.9667	254.6465	8494.092	3.578	3.974e+06	3.814	16.5	
4P _{1/2} – 6S _{1/2}	691.2988	691.1081	433.6655	14465.525	0.903	2.500e+06	1.959	20.3	
4P _{3/2} – 7S _{1/2}	580.3375	580.1766	516.5830	17231.353	0.673	2.348e+06	0.965	35.0	
4P _{3/2} – 8S _{1/2}	534.1173	533.9687	561.2858	18722.481	0.444	1.311e+06	0.612	32.9	
4P _{1/2} – 7S _{1/2}	578.4003	578.2400	518.3131	17289.063	0.476	1.186e+06	0.965	17.7	
4S _{1/2} – 5P _{3/2}	404.5285	404.4142	741.0911	24720.139	0.388	1.150e+06	1.195	16.8	404.5 nm blue line: narrow-line
4S _{1/2} – 5P _{1/2}	404.8356	404.7213	740.5288	24701.382	0.265	1.070e+06	1.157	15.8	404.8 nm blue line
4P _{3/2} – 9S _{1/2}	510.0621	509.9200	587.7568	19605.455	0.325	8.064e+05	0.361	31.8	
4P _{1/2} – 8S _{1/2}	532.4759	532.3279	563.0160	18780.191	0.314	6.620e+05	0.612	16.6	
4P _{3/2} – 6D _{5/2}	536.1067	535.9576	559.2030	18653.004	0.471	4.860e+05	0.179	45.1	

λ and ν from NIST level energies (centre of gravity, no hyperfine structure); λ air uses standard air (Ciddor).
Γ/2π: natural linewidth of the upper level, 1/(2πτ). Branch: share of the upper level's decays that takes this line.

Rydberg excitation wavelengths (vacuum, nm)

Lower → series	n = 30	n = 40	n = 50	n = 70	n = 100	n → ∞	d ₇₀ ARC
4S _{1/2} → nP _{3/2}	286.757	286.246	286.019	285.826	285.727	285.634	0.0011
4P _{3/2} → nS _{1/2}	458.187	456.825	456.227	455.725	455.468	455.230	0.0087
4P _{3/2} → nD _{5/2}	457.819	456.676	456.152	455.698	455.459	455.230	0.0133
5S _{1/2} → nP _{3/2}	722.223	718.989	717.556	716.346	715.722	715.141	0.0039
3D _{5/2} → nP _{3/2}	749.737	746.253	744.708	743.406	742.734	742.108	0.0079
3D _{3/2} → nF _{7/2}	748.888	745.906	744.534	743.343	742.713	742.108	0.0555
5P _{3/2} → nS _{1/2}	985.427	979.149	976.402	974.107	972.932	971.848	0.0175
5P _{3/2} → nD _{5/2}	983.722	978.461	976.058	973.985	972.891	971.848	0.0161
4D _{3/2} → nP _{3/2}	1337.683	1326.632	1321.759	1317.661	1315.551	1313.588	0.0205
4D _{5/2} → nF _{7/2}	1334.983	1325.536	1321.209	1317.465	1315.485	1313.588	0.0898
6S _{1/2} → nP _{3/2}	1347.350	1336.139	1331.196	1327.039	1324.899	1322.908	0.0083
4F _{7/2} → nD _{5/2}	1479.780	1467.907	1462.506	1457.855	1455.407	1453.074	0.0176
6P _{3/2} → nS _{1/2}	1706.393	1687.655	1679.510	1672.731	1669.272	1666.082	0.0287
6P _{3/2} → nD _{5/2}	1701.288	1685.613	1678.495	1672.372	1669.151	1666.082	0.0173

Rydberg term energies from ARC quantum defects + NIST ionisation limit, lower level from NIST. d₇₀ = $| \langle J || e r || J' \rangle |$ to n = 70 in ea0 (calculated).
S → nP values come from a near-cancelling integral: treat them as an order of magnitude.

⁴⁰K hyperfine structure (*I* = 4)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
4S _{1/2}	-285.7308(24)	-	F=7/2: +714.33 F=9/2: -571.46
4P _{1/2}	-34.523(25)	-	F=7/2: +86.31 F=9/2: -69.05
4P _{3/2}	-7.585(10)	-3.44(9)	F=5/2: +55.20 F=7/2: +31.02 F=9/2: -2.29 F=11/2: -46.37
3D _{5/2}	0.71(4)	0.8(8)	F=3/2: -8.48 F=5/2: -6.94 F=7/2: -4.70 F=9/2: -1.66 F=11/2: +2.29 F=13/2: +7.30
3D _{3/2}	1.070(20)	0.40(10)	F=5/2: -7.83 F=7/2: -4.36 F=9/2: +0.36 F=11/2: +6.52
5P _{1/2}	-12.0(9)	-	F=7/2: +30.00 F=9/2: -24.00
5P _{3/2}	-2.450(20)	-1.16(22)	F=5/2: +17.80 F=7/2: +10.03 F=9/2: -0.72 F=11/2: -14.99

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).